

Norbert Frydrysiak

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EDUCATION

Warsaw University of Technology

Warsaw, Poland

Master of Science in Engineering, Data Science

02.2026 – Present

- Relevant Coursework: Advanced Machine Learning, Deep Learning, Cloud Computing, Transformers with PyTorch

Warsaw University of Technology

Warsaw, Poland

Bachelor of Science in Engineering, Data Science

10.2022 – 02.2026

- GPA: 4.81/5.0
- Thesis: *Adapting the gips library for classification problem utilizing discriminant analysis – gipsDA*
- Relevant Coursework: Automated Machine Learning, Stochastic Processes, Methods of Computational Statistics, Computational Intelligence Methods, Financial Mathematics

EXPERIENCE

Teaching Assistant

02.2026 – Present

Warsaw University of Technology

Warsaw, Poland

- Conducting lab classes in "Data Warehousing & Business Intelligence Systems" and "Structured Data Processing"
- Teaching two 15-person groups of 1st and 3rd-year Data Science students for a total of 6 hours per week

Risk Model Validation Intern

07.2025 – 08.2025

Velo Bank S.A.

Warsaw, Poland

- Co-developing an AutoML package in Python for monitoring machine learning models
- Implementing modules for model stress testing, visualizations, and performance analysis in production environments
- Collaborating within a 3-person team, utilizing Git for version control and repository management

Data Science Intern in the Property and Personal Insurance Tariff Team

07.2024 – 09.2024

Generali Poland

Warsaw, Poland

- Preprocessing and analyzing travel insurance claims data, including outlier detection, missing value imputation, and dimensionality reduction
- Developing supervised learning models for claim frequency and average damage utilizing Generalized Linear Models with Poisson and Gamma distributions

PROJECTS

gipsDA (Bachelor's Thesis) | *R, Discriminant Analysis, Statistical Modeling, Git*

- Developed **gipsDA**, an R software package adapting the *gips* library for classification problems utilizing discriminant analysis
- Authored comprehensive **package documentation** and implemented rigorous unit tests to ensure software reliability and mathematical accuracy
- Maintained GitHub repositories for the software implementation and the thesis research, managing datasets, manual tests, and the final thesis document

Transformers Speech | *Python, PyTorch, CNNs, Wav2Vec 2.0, Audio Processing*

- Investigated deep learning architectures for keyword spotting on the TensorFlow Speech Commands dataset, training custom CNN and Audio Transformer models from scratch
- Evaluated and compared the performance of custom models against state-of-the-art transfer learning architectures, including Wav2Vec 2.0
- Implemented audio augmentation techniques and hierarchical cascade classification to effectively handle challenging minority classes and optimize accuracy

TECHNICAL SKILLS

Languages: Python, R, SQL, Java, C/C++, LaTeX, Bash

Data Science & ML: NumPy, Pandas, Scikit-Learn, Scikit-Optimize, XGBoost, PyTorch, SciPy, Matplotlib, Seaborn

Tools & Technologies: Git, Docker, Jupyter, Linux (Debian, Arch), Neovim, REST, Web Scraping

Knowledge & Concepts: Machine Learning, Reinforcement Learning, Stochastic Processes, Statistics, Linear Algebra